D 110213 **(Pages : 3)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2024

Mathematics

MTS 5D 01—APPLIED CALCULUS

(2020 Admission onwards)

Time : Two Hours Maximum : 60 Marks

Section A

Answer any number of questions.
Each question carries 2 *marks.*
Ceiling is 20.

1. Find the domain and range of the function $f(x) = \frac{1}{x-3}$.

2. Find all points of intersection of the graphs of $f(x) = x^2$ and $g(x) = 5x - 6$.

3. Find the slope of the line joining the points $(-2, 5)$ and $(3, -1)$.

4. Differentiate the polynomial $y = 5x^3 - 4x^2 + 12x - 8$.

5. Find $\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$.

6. State Fundamental Theorem of Calculus.

7. Solve for x : $\log_{64} 16 = x$.

8. Expand $\log_2 \frac{y^5}{x^2}$.

9. Find $\int e^{-3x}\, dx$.

Turn over

10. Find $\ln \sqrt{ab}$ if $\ln a = 3$ and $\ln b = 7$.

11. Find the second derivative of $y = 2x^3 - 5x^2 + 2x + 1$.

12. Verify that $\mathrm{F}(x) = \frac{1}{3}x^3 + 5x + 2$ is an antiderivative of $x^2 + 5$.

Section B

Answer any number of questions.
Each questions carries 5 *marks.*
Ceiling is 30.

13. Market research indicates that consumers will buy x thousand units of a particular kind of coffee maker when the unit price is $p(x) = -0.27x + 51$ dollars. The cost of producing the x thousand units is $\mathrm{C}(x) = 2.23x^2 + 3.5x + 85$ thousand dollars. Find the revenue and profit functions, $\mathrm{R}(x)$ and $\mathrm{P}(x)$, for this production process.

14. For what value of the constant A is the following function

$$f(x) = \begin{cases} Ax + 5, & x < 1, \\ x^2 - 3x + 4, & x \geq 1, \end{cases}$$ continuous for all real x ?

15. Find the fifth derivative of $f(x) = \frac{1}{x}$.

16. Determine intervals of concavity for the function $f(x) = 2x^6 - 5x^4 + 7x - 3$.

17. Find the function $f(x)$ whose tangent has slope $3x^2 + 1$ for each value of x and whose graph passes through the point $(2, 6)$.

18. Evaluate $\int_{\frac{1}{4}}^{2} \frac{\ln x}{x}\, dx$.

19. Find the area of the region enclosed by the line $y = 4x$ and the curve $y = x^3 + 3x^2$.

Section C

Answer any **one** *question.*
The questions carries 10 *marks.*

20. Find :

(a) $\lim_{x \to \infty} \frac{x^2}{2x^2 + x + 1}$.

(b) $\lim_{x \to 1} \frac{x^2 - 4x - 5}{x^2 + 4x + 3}$.

21. Evaluate :

(a) $\int e^{5x+2}\, dx$.

(b) $\int_0^1 8x \left(x^2 + 1\right)^3 dx$.

(1 × 10 = 10 marks)